

Growa Platform — Intellectual Property Vault

Version: 1.0

Date: September 2026

Purpose: Formal record of proprietary software architecture, methods, and product design without exposing credentials, infrastructure secrets, or security-sensitive implementation details.

About This Vault

This knowledge base documents the **Growa** agricultural operations platform and its first sovereign deployment, **Growa Qatar**. It is structured like an **Obsidian** vault: interconnected notes, clear hierarchy, and cross-references.

What this vault establishes (IP deposit):

- Platform architecture and design patterns
- Product capabilities and analytical methods
- Data models at conceptual level
- AI briefing methodology
- Security and governance approach (high level)

What this vault deliberately excludes:

- API keys, credentials, and environment secrets
- Production URLs, project identifiers, and infrastructure endpoints
- Exact database security policy logic
- Internal admin tooling and impersonation mechanics
- Partner-specific operational data

Navigation Map

Platform Foundation

- 01-Executive-Summary — One-page positioning and value proposition

- 02-Platform-Architecture — Platform → Country → Organization hierarchy
- 03-Technology-Stack — Languages, frameworks, and infrastructure patterns

Product & Capabilities

- 04-Module-Registry — Declarative workspace module system (core IP)
- 05-Intelligence-Workspaces — Water, energy, analytics, weather
- 06-Map-Centric-UX — Satellite map as national operating surface
- 07-Compliance-Governance — Inspections, cases, alerts, collaboration
- 08-Supply-Chain — Logistics and procurement visibility

Technical IP

- 09-Data-Model — Conceptual entity relationships
- 10-Auth-Access-Model — Identity, roles, and permission layers
- 11-Row-Level-Security — Data isolation strategy (conceptual)
- 12-Growa-AI-Methodology — Grounded government briefing system
- 13-Internationalization — Bilingual EN/AR architecture
- 14-API-Design — Server-side BFF pattern

Legal & Deposit

- 15-IP-Claims-Summary — Patent/trade-secret positioning
- 16-Security-Redaction-Policy — What was excluded and why
- 17-Glossary — Canonical terminology

Ownership

Growa Platform and **Growa Qatar** are proprietary software.

Rights reserved by the authorized deploying institution(s) in the State of Qatar.

Tags

#growa #ip-deposit #agriculture #sovereign-platform #qatar

Executive Summary

What Growa Is

Growa is a multi-country agricultural operations platform designed for sovereign government deployments. It unifies maps, operational data, resource intelligence, compliance workflows, supply visibility, and AI-assisted briefings in a single secure environment.

Growa Qatar is the first country deployment — a national digital operating system for Qatar's agricultural sector, customized with bilingual support (English and Arabic), national geography, and institution-specific workspaces.

The Problem It Solves

Agricultural governance involves many actors: ministries, inspection teams, state food operators, financial institutions, and farm companies. Each needs visibility into production, water use, energy consumption, compliance, and supply — but data is often fragmented across spreadsheets, paper records, and disconnected systems.

Growa answers:

How can authorized institutions see the full agricultural picture, act together, and protect national food security?

Core Value Propositions

Dimension	Value
Spatial intelligence	Map-first UX connects every metric to geographic context
Resource accountability	Water and energy intensity measured per crop and per producer
Sovereign control	Each country deployment is isolated with its own data boundary
Multi-tenant governance	Organizations see only authorized data; collaboration is explicit
Grounded AI	Briefings cite live platform data; no autonomous decision-making
Bilingual by design	Full English/Arabic with RTL layout support

Target Users

- Government ministries (national oversight, policy, inspections)
 - Field inspection teams (cases, evidence, corrective actions)
 - State food operators (production and supply monitoring)
 - Financial institutions (agricultural program oversight)
 - Farm companies (site operations and resource tracking)
 - Research and technical support bodies
-

Platform in One Sentence

Growa Qatar is the national command centre for agricultural operations — connecting maps, weather, water and energy analysis, food security planning, compliance, supply, and intelligence so Qatar can protect its food security today and plan for tomorrow.

Related Notes

- 02-Platform-Architecture
- 04-Module-Registry
- 12-Growa-AI-Methodology
- 15-IP-Claims-Summary

Platform Architecture

Three-Layer Hierarchy

Growa uses a deliberate separation between reusable platform code and country-specific deployment:

```
Growa Platform (Base)
└─ Growa Qatar (Country Deployment)
    ├── Government Ministry (Organization)
    │   └─ Departments (Food Security, Inspection, etc.)
    ├── State Food Operator (Organization)
    ├── Financial Institution (Organization)
    └─ Farm Companies (Organizations)
```

Layer 1: Growa Platform (Base)

The shared foundation reused across future country deployments:

- Core application architecture and codebase
- Design system and component library
- Authentication and authorization framework
- Operational ontology (farm, greenhouse, sensor, alert, cycle, etc.)
- Module registry and navigation patterns
- AI briefing pipeline architecture
- Code conventions and extension points

Intentionally country-agnostic:

- No branding or legal labels
- No map data or geographic boundaries
- No organization-specific data
- No country-specific role presets beyond templates

Layer 2: Country Deployment (Growa Qatar)

Each sovereign deployment customizes:

Element	Qatar Customization
Branding	National colors, bilingual labels
Legal labels	State terminology, ministry names
Organizations	Authorized national entities
Regions	National municipalities and geography
Map layers	Country satellite and operational overlays
Role presets	Government structure templates
Language	Arabic and English

Isolation principle: Each future deployment (e.g., other GCC countries) receives its own isolated database instance. No cross-country data sharing at infrastructure level.

Layer 3: Organizations

Within a country deployment, multiple authorized organizations operate:

Organization Type	Description
<code>government_master</code>	Top-level government authority
<code>government</code>	Government departments and agencies
<code>farm_company</code>	Licensed agricultural operators
<code>private</code>	Approved private entities
<code>public</code>	Public-sector operators

Key Principles

1. **Multi-actor, not ministry-only** — Platform serves diverse institution types
2. **Organization scoping** — Users belong to organizations with defined roles
3. **Data isolation** — Organizations see own data unless explicit cross-org sharing
4. **Shared infrastructure** — Common deployment, isolated tenant data

Access Model

```
Country Instance
├─ Organization
│   └─ Department (optional)
│       └─ User Membership
│           └─ Role (permission template)
│               └─ Scope (region, farm, object boundaries)
```

See 10-Auth-Access-Model for permission resolution.

Application Architecture Pattern

Browser (React / Next.js App Router)
├─ Map-centric dashboard shell
├─ Module workspaces (slide-over panels)
└─ Auth + i18n context providers
Server Layer (Next.js API Routes / BFF)
├─ Operations APIs (farms, map points, insights)
├─ Weather proxy (keys never exposed to client)
└─ AI analysis endpoint (server-side only)
Data Layer (PostgreSQL via managed backend)
├─ Auth service (identity, sessions)
├─ Row Level Security (tenant isolation)
└─ RPC functions (complex authorization)

See 14-API-Design and 11-Row-Level-Security.

Deployment Model

Environment	Purpose
Local	Developer workstations
Preview	Staging and QA
Production	Sovereign national deployment

Configuration is environment-driven via public-safe feature flags. Secrets are injected at deployment time and never committed to source control.

Related Notes

- [03-Technology-Stack](#)
- [04-Module-Registry](#)
- [09-Data-Model](#)
- [10-Auth-Access-Model](#)

Technology Stack

Frontend

Component	Technology	Role
Framework	Next.js 16 (App Router)	Server and client rendering, routing, API routes
UI Library	React 19	Component model
Language	TypeScript (strict)	Type safety across codebase
Styling	Tailwind CSS 4	Utility-first responsive design
Components	shadcn/ui + Radix UI	Accessible, composable UI primitives
Maps	Leaflet	Interactive satellite and operational map
Charts	Recharts	Analytics visualizations
Forms	React Hook Form + Zod	Validated form handling
Themes	next-themes	Light/dark mode support

Backend & Data

Component	Technology	Role
Database	PostgreSQL	Relational data store
Backend service	Managed auth + database platform	Identity, realtime, storage
Auth	SSR-compatible auth client	Session cookies, JWT refresh
Migrations	SQL migration files	Versioned schema evolution
Security	Row Level Security (RLS)	Database-enforced tenant isolation

AI & External Services

Component	Technology	Role
LLM	Google Gemini (server-side)	Government briefing generation
Weather	External weather API (proxied)	Grid-based agronomic data
Analytics	Vercel Analytics	Usage telemetry (optional)

Security pattern: All external API keys are held server-side only. Client never receives provider credentials.

Internationalization

Component	Implementation
Locales	English (<code>en</code>), Arabic (<code>ar</code>)
Direction	Automatic RTL for Arabic
Storage	Browser localStorage persistence
Dictionaries	Flat key-value translation maps

See 13-Internationalization.

Development Tooling

- Package manager: pnpm
- Linting: ESLint with Next.js config
- Type generation: Database types synced from schema

Architectural Conventions

1. **Server Components first** — Initial data fetched server-side where possible
2. **BFF API routes** — External services called only from server
3. **Context providers** — Theme → Auth → i18n composition at app root
4. **Feature flags** — MFA, SSO, realtime toggled via environment
5. **Idempotent migrations** — Defensive schema evolution for mixed DB states

Related Notes

- [14-API-Design](#)
- [11-Row-Level-Security](#)
- [12-Growa-AI-Methodology](#)

Module Registry — Core IP Artifact

Overview

The **Module Registry** is a declarative, code-first catalog of every workspace module in the platform. It is a central intellectual property artifact: it encodes product intent, role-aware navigation, visibility rules, and UX blueprints in a single authoritative source.

Unlike ad-hoc route tables, the registry describes **what each module is for, who can see it, and what actions it supports** — before backend data is wired.

Module Definition Schema

Each module entry contains:

Field	Purpose
<code>id</code>	Stable module identifier (e.g., <code>national-overview</code>)
<code>label</code>	Display name, with optional role-specific overrides
<code>purpose</code>	One-line mission statement
<code>defaultContent</code>	Expected content when module loads
<code>allowedActions</code>	Permitted user actions in this module
<code>visibilityScope</code>	Org types, permissions, and data-layer requirements
<code>submenu</code>	Role-aware navigation items within the module
<code>icon</code>	Visual identifier for sidebar

Visibility Scope Model

Modules are shown only when all visibility conditions pass:

```
interface ModuleVisibilityScope {
  allowedOrgTypes: OrgType[] | '*' // Which organization types
  requiredPermissions: PermissionFlag[] // Required capability flags
  requiredLayerVisibility?: { // Shared data layer access
    regulatory?: VisibilityLevel[]
    commercial?: VisibilityLevel[]
    finance?: VisibilityLevel[]
    technical_support?: VisibilityLevel[]
  }
}
```

Visibility levels: FULL , SUMMARY , APPROVAL , NO

This enables fine-grained control: a module may require regulatory layer access at SUMMARY or above, while another requires commercial FULL access.

Role-Aware Labels and Submenus

The registry supports **role-specific UX** without duplicating module definitions:

- **Labels:** ministry_inspector may see "Alerts & Incidents" where ministry_admin sees "Alerts & Risks"
- **Submenus:** Inspector role gets inspection-focused map views; admin gets national map views

This pattern reduces navigation sprawl while preserving role-appropriate language.

Role Menu Blueprints

Predefined navigation layouts per ministry role profile:

Ministry Admin Blueprint

- **Landing:** National Overview
- **Primary:** National Overview, Live Map, Monitoring, Alerts, Compliance, Production & Harvest
- **Secondary:** Inter-Agency Collaboration, Programs & Policy, Reports, Support, Settings

Ministry Inspector Blueprint

- **Landing:** Inspection Dashboard
- **Primary:** Inspection Dashboard, Live Map, Compliance Cases, Non-Conformities, Corrective Actions, Alerts

- **Secondary:** Farms & Sites, Evidence & Attachments, Reports, Support, Settings

Registered Modules (Catalog)

Module ID	Purpose
<code>national-overview</code>	Sovereign executive situational awareness
<code>inspection-dashboard</code>	Inspector workload and queue management
<code>live-map</code>	Primary spatial decision surface
<code>monitoring</code>	Environmental and data health signals
<code>alerts-center</code>	Risk triage, escalation, and closure
<code>compliance-inspections</code>	Regulatory posture and inspection planning
<code>production-harvest</code>	Production readiness and harvest risk
<code>inter-agency-collaboration</code>	Cross-agency case and data exchange
<code>programs-policy</code>	Public program and policy tracking
<code>reports-center</code>	Institutional reporting packages
<code>compliance-cases</code>	End-to-end compliance case lifecycle
<code>non-conformities</code>	Violation severity and recurrence tracking
<code>corrective-actions</code>	Corrective action execution monitoring
<code>farms-sites</code>	Institutional farm and site registry views
<code>evidence-attachments</code>	Inspection evidence lifecycle
<code>support</code>	Institutional help and service requests
<code>settings</code>	Workspace configuration

Navigation Resolution Algorithm

When a user signs in, navigation is resolved through a priority chain:

1. **Code registry** — Ministry role blueprints filtered by visibility
2. **Database menus** — Per-role JSON navigation stored in `role_navigation` table
3. **Fallback** — Default minimal menu for unassigned roles

Each candidate module passes:

- Organization type check
 - Permission flag check
 - Shared layer visibility check
-

Route Pattern

Modules use query-parameter routing to avoid 404s for in-progress features:

```
/dashboard?module=national-overview  
/dashboard?module=live-map
```

Dedicated routes exist for mature features:

```
/dashboard/farms  
/dashboard/supply-overview  
/dashboard/settings
```

See 06-Map-Centric-UX.

IP Significance

The Module Registry represents a **product ontology** — a machine-readable description of government agricultural software capabilities. It enables:

- Consistent role-based UX across deployments
 - Country-specific customization without rewriting navigation
 - Auditability of who can access what capability
 - Progressive implementation (shell UI before backend wiring)
-

Related Notes

- 05-Intelligence-Workspaces
- 07-Compliance-Governance
- 10-Auth-Access-Model
- 15-IP-Claims-Summary

Intelligence Workspaces

Overview

Growa provides dedicated analytical workspaces that measure agricultural performance across water, energy, production, and weather dimensions. These workspaces share a common metrics pipeline and connect to the Growa AI briefing layer.

Metrics Pipeline

```
Map polygons (custom_point_polygons)
    ↓
Farm crop insights (per-point production/water/energy)
    ↓
Workspace aggregates (national KPIs, rankings)
    ↓
Growa analysis context (structured digest)
    ↓
AI briefing (optional)
```

Data Analytics Workspace

Purpose: Cross-resource performance view combining production, water, and energy.

Analytical Outputs

Metric	Description
Total production	Estimated tons across monitored network
Resource intensity	Combined water + energy per ton of output
Efficiency scores	Per-area and per-producer performance
Crop matrix	Farms, output, resources, and scores by crop
Producer rankings	Efficiency, volume, and intensity leaders
Data gaps	Crops or areas with incomplete monitoring

User Actions

- Compare crop portfolio performance
 - Identify top and at-risk producers
 - Navigate from table rows to map locations
 - Request executive briefings via Growa AI
-

Water Intelligence Workspace

Purpose: Measure water consumption relative to agricultural output.

Analytical Outputs

Metric	Description
Total water consumption	Cubic metres across monitored areas
Water intensity	m ³ per ton of crop production
Irrigation pressure	Water share of total resource use
Crop breakdown	Water use and intensity by crop type
Producer rankings	Farms ranked by water efficiency
Quality scores	Linked to monitored production polygons

Policy Questions Answered

- Which crops consume disproportionate water for their output?
 - Which producers need conservation support or inspection?
 - Where is national water use concentrated?
-

Energy Intelligence Workspace

Purpose: Track energy consumption across agricultural operations.

Analytical Outputs

Metric	Description
Total energy consumption	kWh across monitored operations
Energy intensity	kWh per ton of production
Average per site	Energy use per farm or facility
Crop breakdown	Energy demand by crop type
Producer rankings	Efficiency leaders and laggards

Policy Questions Answered

- Which operations have unusually high energy demand?
- How does energy input relate to food output nationally?
- Where should efficiency programmes focus?

Weather Workspace

Purpose: National grid weather with agronomic risk indicators.

Coverage

- Entire national territory via configurable grid
- Current conditions and historical trends per grid cell

Analytical Domains

Domain	Insight
Light & solar radiation	Greenhouse and growth planning
Temperature & humidity	Heat stress and moisture
Wind	Spraying safety, structural risk
Rainfall	Recent precipitation windows
Irrigation indicators	Evapotranspiration, water stress
Agronomic risk	Disease risk, thermal stress indices

User Actions

- Select any grid cell for environmental profile
- Compare weather trends over time
- Plan irrigation and field operations proactively

RSS Information Feed

Curated external news and information alongside the operational map — agricultural news, market signals, and policy context without leaving the workspace.

Shared UI Patterns

All intelligence workspaces share:

- Slide-from-left panel over the persistent map
 - Consistent KPI cards and data tables
 - Map navigation from data rows to geographic locations
 - Optional Growa AI briefing panel with preset prompts
-

Related Notes

- 06-Map-Centric-UX
- 12-Growa-AI-Methodology
- 09-Data-Model
- 08-Supply-Chain

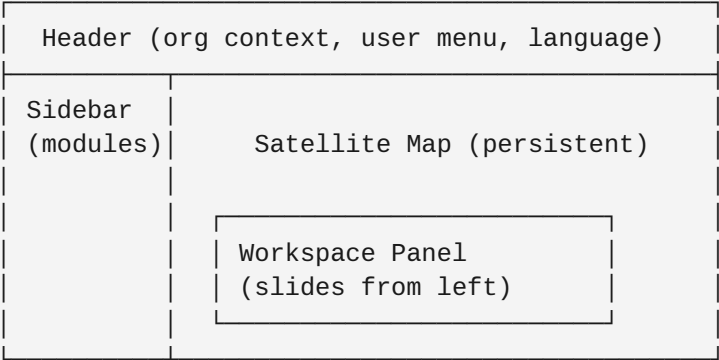
Map-Centric User Experience

Design Philosophy

Agriculture is inherently spatial. Growa treats the **interactive satellite map** as the persistent operating surface — not a secondary widget. All analytical workspaces slide over the map, preserving geographic context while presenting data.

Every number on a dashboard should lead to a place on the ground.

Map Shell Architecture



Map Capabilities

Capability	Description
National → site zoom	From country view to individual farm
Layer presets	Toggle operational overlays
Search & locate	Find farms, facilities, points
Saved views	Persist operational map configurations
Custom map points	User-drawn markers (farm, facility, sensor)
Polygon overlays	Operational areas with crop and resource metrics
Scope switching	National, regional, or inspection-focused views

Custom Map Objects

Map Points

User-placed markers representing farms, facilities, sensors, or other operational sites. Bilingual labels supported.

Polygons

Drawn operational areas linked to:

- Crop type assignment
- Production, water, and energy totals
- Performance scores
- External reference links (when authorized)

Crop Insights

Per-point analytical records aggregating production and resource metrics — the foundation for intelligence workspace calculations.

Navigation Integration

- Module selection via `?module=` query parameter keeps map mounted

- Clicking a data table row pans/zooms map to corresponding location
 - Inspector role receives inspection-focused map submenu items
 - Admin role receives national and regional map views
-

UX IP Claims

1. **Persistent map shell** — Analytical panels are overlays, not page replacements
 2. **Bidirectional data-map linking** — Tables and map are always connected
 3. **Role-aware map modes** — Same map, different operational lenses per role
 4. **Progressive module loading** — Registry-defined shells render before full backend wiring
-

Related Notes

- 04-Module-Registry
- 05-Intelligence-Workspaces
- 09-Data-Model

Compliance & Governance Workspaces

Overview

Government and inspection teams use dedicated workspaces to enforce agricultural standards, manage compliance cases, and coordinate regulatory action across Qatar's agricultural sector.

These modules are defined in the 04-Module-Registry and gated by the `regulatory` shared data layer.

Inspection Dashboard

Audience: Field inspectors

Purpose: Daily workload management

Capabilities

- Personal inspection queue
 - Items due this week
 - High-priority cases
 - Assigned regions
 - Open findings tracker
 - Completion status metrics
-

Compliance & Inspections

Audience: Regulatory leadership

Purpose: Institutional compliance posture

Capabilities

- Inspection planning and scheduling
- Non-conformity pipeline
- Corrective action tracking
- Compliance score trends
- Regulatory report generation

Compliance Cases

Purpose: End-to-end case lifecycle management

Case States

- Open → In Review → Awaiting Evidence → Escalated → Closed

Capabilities

- Case queue grouped by state
 - Evidence completeness tracking
 - Escalation workflows
 - Map-linked case locations
-

Non-Conformities

Purpose: Track violation patterns

Analytical Views

- Open non-conformities by severity
 - Geographic distribution (by region, by farm)
 - Repeat offender identification
 - Severity breakdown trends
-

Corrective Actions

Purpose: Monitor remediation execution

Queues

- Open actions
- Overdue actions
- Pending verification
- Closed actions

Ensures identified problems are actually resolved, not just documented.

Evidence & Attachments

Purpose: Documentary trail for compliance decisions

Capabilities

- Media uploads (photos, documents)
 - Evidence library
 - Case-linked attachments
 - Missing evidence alerts
 - Signed report storage
-

Alerts & Risk Management

Purpose: Centralised risk triage

Coverage

- Active operational and compliance alerts
- Critical incidents
- Risk hotspots (geographic concentrations)
- Escalation tracking
- Resolved vs. open trend analysis

Actions

- Acknowledge alerts
 - Escalate to appropriate authority
 - Link alerts to compliance cases and map locations
-

Inter-Agency Collaboration

Purpose: Structured cooperation between authorized agencies

Capabilities

- Shared cross-organization cases

- Data sharing requests with approval workflow
- Visibility rules (full, summary, approval-gated)
- Collaboration audit log

Prevents both data silos and uncontrolled informal sharing.

Programs & Policy Tracking

Purpose: Monitor public agricultural programmes

Analytical Views

- Programme enrollment status
 - Regional coverage gaps
 - Programme exceptions
 - Incentive monitoring
 - Policy performance against objectives
-

Reports Center

Purpose: Institutional reporting for leadership and regulators

Report Types

- Executive summaries
 - Compliance and inspection reports
 - Water use reports
 - Production and harvest reports
 - Programme performance reports
 - Inspector-specific regional summaries
-

Related Notes

- 04-Module-Registry
- 10-Auth-Access-Model
- 09-Data-Model

Supply Chain & Logistics

Overview

For organizations responsible for sourcing, procurement, and distribution, the supply overview workspace monitors the flow of goods from origin to destination — closing the loop between field production and market availability.

Strategic Context

Food security depends not only on what a nation grows, but on what it sources, imports, and distributes. Supply analysis complements production intelligence with logistics visibility.

Key Performance Indicators

KPI	Description
Available contract volume	Supply contracted and ready
Goods in transit	Volumes currently moving through chain
At-risk deliveries	Shipments facing timeline or disruption risk
Average lead times	Origin-to-delivery duration trends

Supply Flow Records

Individual commodity flows tracked with:

- Commodity type
 - Origin and destination
 - Current status
 - Estimated arrival
 - Risk flags
-

Action Queue

Open sourcing actions requiring attention — enabling proactive intervention before shortages materialize.

Data Model (Conceptual)

Entity	Purpose
supply_overview_snapshots	Daily KPI aggregates
supply_flows	Individual commodity movement records
supply_action_queue	Open sourcing actions

See 09-Data-Model.

User Experience

- Dedicated route: /dashboard/supply-overview
 - Tailored for state food operator organization types
 - Connects sourcing decisions with production outlook data from field intelligence
-

Related Notes

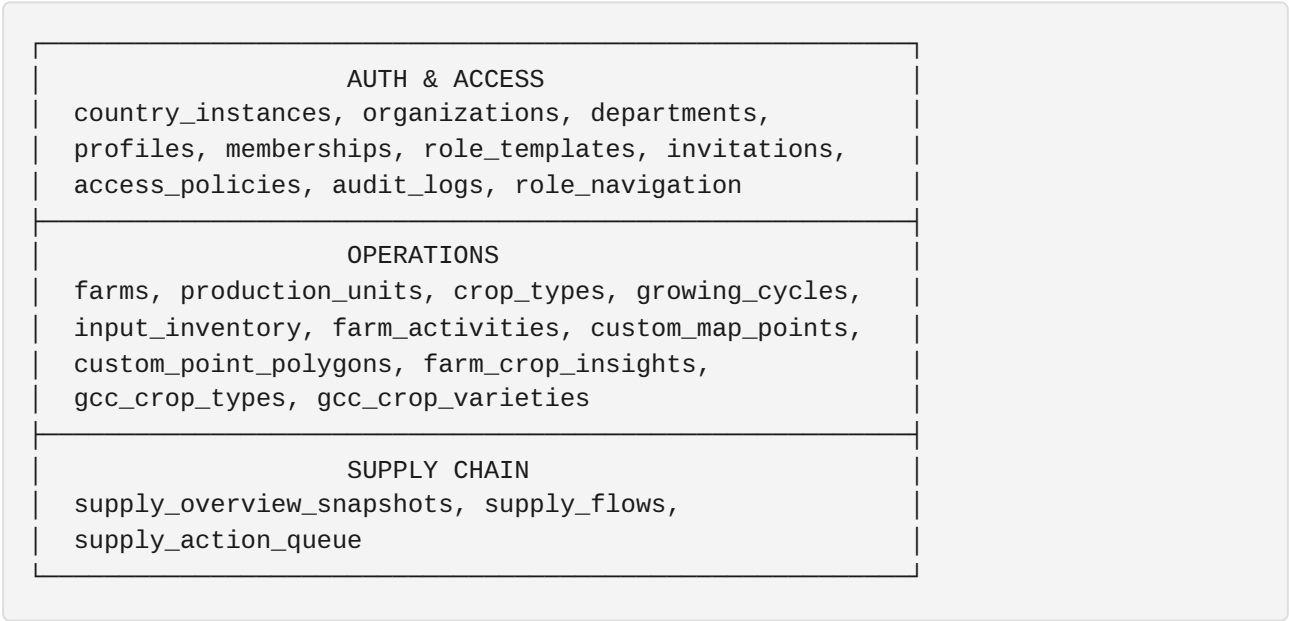
- 05-Intelligence-Workspaces
- 09-Data-Model
- 01-Executive-Summary

Data Model — Conceptual Overview

Design Principles

- 1. **Organization-scoped tenancy** — All operational data belongs to an organization
 - 2. **Bilingual by default** — Names and descriptions in English and Arabic
 - 3. **Geographic linkage** — Entities connect to map coordinates and regions
 - 4. **Audit readiness** — Significant actions logged for compliance
 - 5. **Defensive evolution** — Schema migrations are idempotent for mixed deployment states
-

Domain Map



Auth & Access Domain

Entity	Purpose
country_instances	Country deployment container
organizations	Multi-tenant actors (government, farm company, etc.)
departments	Subdivisions within organizations
regions	Geographic hierarchy (municipalities)
profiles	Application user profile (1:1 with identity)
user_organization_members	Active membership with role assignment
memberships	Blueprint membership with scope assignments
role_templates	Predefined permission templates
permission_templates	Granular domain/action permissions
invitations	Invitation-only onboarding records
access_policies	Organization security policies
role_navigation	Per-role menu configuration (JSON)
role_delegations	Temporary role delegation
audit_logs	Security and compliance audit trail

Operations Domain

Entity	Purpose
farms	Registered agricultural sites
production_units	Fields, pens, tanks within farms
crop_types	Reference crop taxonomy
livestock_types	Livestock reference data
aquaculture_species	Aquaculture reference data
growing_cycles	Crop production lifecycle
livestock_batches	Livestock production batches
aquaculture_cycles	Aquaculture production cycles
input_inventory	Agricultural input stock
farm_activities	Operational activity log
custom_map_points	User-drawn map markers
custom_point_polygons	Polygon overlays with metrics and scores
custom_crop_types	User-defined crop labels
gcc_crop_types	GCC regional crop catalog
gcc_crop_varieties	Crop variety catalog
farm_crop_insights	Per-point production/water/energy totals

Supply Chain Domain

Entity	Purpose
supply_overview_snapshots	Daily supply KPI aggregates
supply_flows	Individual commodity flow records
supply_action_queue	Open sourcing actions

Key Relationships

Organization —< Farms —< Production Units
Organization —< Custom Map Points —< Polygons —< Crop Insights
Organization —< Supply Flows
User —< Memberships >— Organization
Role Template —< Permission Templates

Enumeration Types

Core enums include:

- Organization types: `government_master` , `government` , `farm_company` , `private` , `public`
- Farm types: crop, livestock, aquaculture, mixed
- Farm status: active, inactive, pending, suspended
- Invitation status: pending, accepted, expired, revoked
- Supply flow status: in-transit, delivered, at-risk, delayed

Shared Data Layers

Cross-cutting visibility domains applied to modules:

Layer	Content Domain
<code>regulatory</code>	Compliance, inspections, water/food security
<code>commercial</code>	Harvest, traceability, supplier eligibility
<code>finance</code>	Farm dossier, KPIs, credit
<code>technical_support</code>	Device health, diagnostics

Visibility levels: `FULL` , `SUMMARY` , `APPROVAL` , `NO`

Related Notes

- 10-Auth-Access-Model
- 11-Row-Level-Security
- 05-Intelligence-Workspaces

Authentication & Access Model

Core Principles

1. **No public signup** — Users cannot self-register
 2. **Invitation-only** — Administrators invite authorized users
 3. **Organization-scoped** — Every user belongs to organizations with roles
 4. **Server-side privilege** — Sensitive operations never exposed to client
 5. **Layered permissions** — Never flattened to a single role string
-

Authentication Methods

Primary: Email + Password

- Managed identity service with secure session cookies
- Short-lived access tokens with refresh rotation
- Email verification before data access

Roadmap: Single Sign-On

- Enterprise SSO for government organizations (Azure AD, etc.)
- Feature-flagged per deployment phase

Roadmap: Multi-Factor Authentication

- TOTP for sensitive roles
 - Mandatory MFA for administrative roles (phased rollout)
-

Invitation Flow

```
Admin initiates invitation
  → Secure token generated (server-side only)
  → Email with activation link
  → User sets password
  → Profile and membership created
  → User can sign in
```

Invitation States

pending → accepted | expired | revoked

Security Controls

- Cryptographically secure single-use tokens
- Configurable expiration (default 72 hours)
- Rate limiting on invitation creation
- Full audit trail

Permission Layer Stack

Effective permissions combine six layers:

```
Layer 1: Country Instance (deployment boundary)
Layer 2: Organization (institution type and membership)
Layer 3: Department (organizational subdivision)
Layer 4: Geographic Scope (region, municipality)
Layer 5: Object Scope (specific farms, sites)
Layer 6: Action Permissions (read, write, approve, admin)
```

Rule: Effective permissions = Role Template + Scoped Assignments

Role Templates (Examples)

Role Code	Scope	Key Capabilities
org_master_admin	Organization-wide	Full admin control
org_admin	Department-scoped	User and workflow management
food_security_director	Country-wide view	Strategic oversight, executive reports
regional_ops_officer	Region-scoped	Regional operations and alerts
agricultural_inspector	Assigned scope	Inspections, evidence, findings
agronomist	Assigned scope	Crop and sensor data advisory
compliance_officer	Organization-wide	Compliance reports and audit
analyst	Assigned scope	Read-only dashboards and exports
executive_readonly	Organization-wide	KPIs without operational detail
external_operator	Own assets only	Restricted partner access

Permission Flags (Application Layer)

Flag	Meaning
canView	Read access to authorized data
canEdit	Create and update records
canManageUsers	Invite and manage team members
canDeleteOrganization	Organization deletion (restricted)
canShareData	Initiate cross-org data sharing
canViewRegulatory	Access regulatory layer data
canViewCommercial	Access commercial layer data
canViewFinance	Access financial layer data
canViewTechnical	Access technical support layer

Mapped from organization type and role in application hooks.

Permission Resolution Algorithm

1. Load user's organization memberships
2. For each membership, load role template permissions
3. Load scope assignments (geographic + object)
4. For each access check:
 - a. Does role template grant the action?
 - b. Does scope include the target entity?
 - c. Does shared layer visibility permit access?
 - d. If all pass → GRANTED; otherwise → DENIED

Session Management

- HTTP-only secure cookies (not accessible via JavaScript)
- Access token expiry with automatic refresh
- Session invalidation on password reset and admin action
- Account states: `active`, `pending`, `suspended`, `revoked`

JWT vs Database

JWT (coarse): Country, organization hint, role hint — for quick eligibility

Database (fine-grained): Full permissions, scopes, memberships — for authorization decisions

Rule: Never embed large permission matrices in JWT tokens.

Related Notes

- 11-Row-Level-Security
- 04-Module-Registry
- 16-Security-Redaction-Policy

Row Level Security Strategy

Overview

Growa enforces data isolation at the **database layer** using PostgreSQL Row Level Security (RLS). This is a defense-in-depth approach: even if application code has a bug, unauthorized data access is blocked by the database itself.

Core Principles

1. **Default deny** — No access unless explicitly granted by policy
 2. **Membership-based** — Access tied to organization membership
 3. **Organization scoping** — Users see only their organization's data
 4. **Service role isolation** — Privileged server operations use elevated role that bypasses RLS
 5. **No client-side trust** — Authorization never depends solely on frontend checks
-

Policy Categories

Category	Approach
Read (SELECT)	User must be member of owning organization
Write (INSERT/UPDATE)	User must have appropriate role within organization
Delete	Restricted to admin roles; soft delete preferred
Cross-org	Explicit sharing agreements required

Membership Verification Pattern

RLS policies verify access through organization membership:

- Check that `auth.uid()` matches an active membership record
- Verify the target row's `organization_id` matches the membership
- Optional geographic scope checks for region-scoped roles

Note: Exact policy predicates are intentionally omitted from this document to prevent security disclosure.

Helper Functions

Database RPC functions support complex authorization without exposing logic to clients:

Function Category	Purpose
Membership checks	Verify user belongs to organization
Visibility resolution	Determine shared layer access level
Effective role	Resolve role with delegation support
Organization creation	Atomic org + owner setup

RLS + Application Layer

```
User Request
  ↓
Middleware (session validation)
  ↓
API Route (permission flag check)
  ↓
Database Query (RLS policy enforcement)
  ↓
Response (only authorized rows returned)
```

Both application-level permission flags AND database RLS must pass.

Audit & Compliance

- All significant data access patterns are auditable
- RLS provides immutable enforcement regardless of client manipulation
- Policy changes are versioned through migration files

Related Notes

- 10-Auth-Access-Model

- 16-Security-Redaction-Policy
- 09-Data-Model

Growa AI Methodology

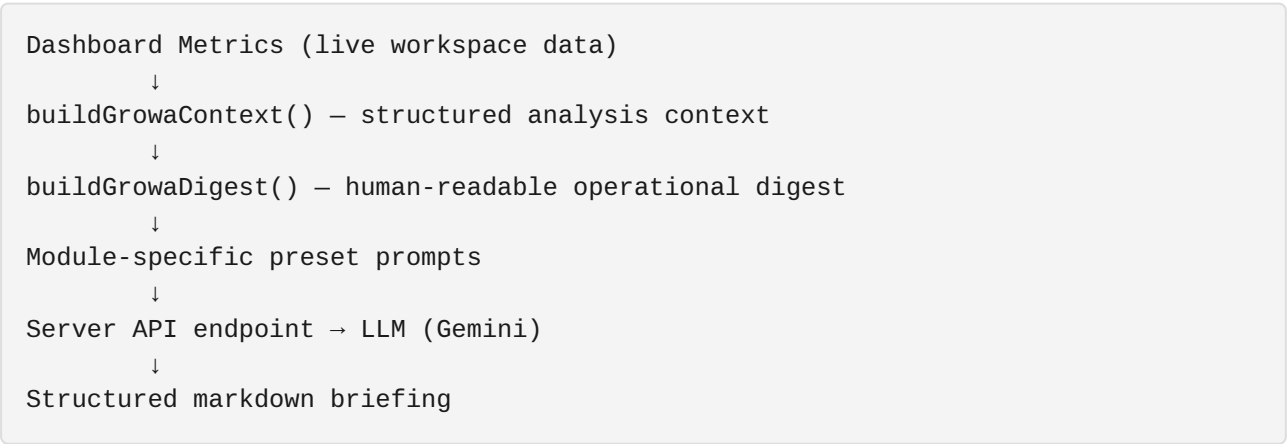
Overview

Growa AI is a grounded government briefing system — not a general-purpose chatbot. It reads live operational data from the platform and produces structured, evidence-based briefings for authorized government users.

Design Constraints

Constraint	Rationale
Grounded in platform data only	Prevents hallucinated farms, crops, or metrics
Server-side execution only	API keys never exposed to browser
Structured output format	Consistent executive briefing structure
English output only	Government briefing standard language
No autonomous decisions	AI accelerates understanding; humans retain authority
Data gap transparency	Explicitly flags missing or incomplete data

Pipeline Architecture



Supported Modules

Module	Briefing Focus
data-analytics	Executive briefing, intervention priorities, food security outlook
water-intelligence	Water policy, conservation priorities, drought resilience
energy-intelligence	Efficiency assessment, grid pressure, decarbonization roadmap

Operational Digest Structure

The digest is the **single source of truth** passed to the LLM:

National Headline

- Tracked crops, producers, and polygons
- Total production, water, and energy
- Average performance scores
- Top and lowest efficiency producers
- Module-specific focus metrics (water intensity, energy intensity, etc.)

Crop Matrix

Per-crop breakdown: production share, water share, energy share, scores, farm count, intensity metrics

Producer Rankings

Top producers by efficiency with production, resource use, and crop portfolio

At-Risk Producers

Low efficiency / high resource pressure flags

Data Alerts

Explicit gaps and anomalies in monitoring coverage

Analysis Context Schema

Structured context object includes:

- `module` — Active workspace identifier
 - `headline` — National KPI aggregates
 - `crops[]` — Per-crop analytical rows
 - `topProducers[]` — Efficiency leaders
 - `atRiskProducers[]` — Flagged underperformers
 - `alerts[]` — Data quality and coverage warnings
 - `generatedAt` — Snapshot timestamp
-

Output Format (5 Sections)

Every briefing follows a fixed structure:

1. **Executive Summary** — Key findings in plain language
 2. **Evidence** — Metrics cited from the operational digest
 3. **Risk Signals** — Outliers, at-risk producers, anomalies
 4. **Recommended Actions** — Concrete government interventions
 5. **KPIs & Data Gaps** — Measurable targets and missing data
-

Anti-Hallucination Rules

Embedded in system prompts:

- Must cite digest metrics by name and value
 - Must not invent farms, crops, or producers not in digest
 - Must flag insufficient data via alerts section
 - Must distinguish observation from recommendation
 - Must not claim regulatory authority or make binding decisions
-

Module-Specific Analysis Frameworks

Each workspace module defines an analytical methodology:

- **Water:** Intensity benchmarking, irrigation pressure, crop-level water accounting
 - **Energy:** kWh/t efficiency, producer rankings, decarbonization pathways
 - **Analytics:** Cross-resource efficiency, portfolio concentration, intervention prioritization
-

UI Integration

`GrowaIntelligencePanel` component in intelligence workspaces provides:

- Preset prompt buttons (one-click briefings)
 - Custom prompt input (within module scope)
 - Markdown-rendered response display
 - Loading and error states
-

IP Significance

The Growa AI methodology represents a novel approach to **government-grade grounded AI**:

1. Pre-computed structured digest before LLM invocation
 2. Module-specific analytical frameworks
 3. Fixed output schema for executive consumption
 4. Explicit anti-hallucination guardrails
 5. Data gap transparency as first-class output
-

Related Notes

- 05-Intelligence-Workspaces
- 14-API-Design
- 15-IP-Claims-Summary

Internationalization (i18n)

Overview

Growa is designed for bilingual operation from the ground up. The Qatar deployment supports **English** and **Arabic** with full right-to-left (RTL) layout switching.

Architecture

Component	Implementation
Locales	<code>en</code> (default), <code>ar</code>
Direction	<code>ltr</code> for English, <code>rtl</code> for Arabic
Storage	Browser <code>localStorage</code> key for persistence
Provider	React context at application root
Toggle	User menu language switcher

Translation API

```
const { locale, direction, setLocale, t } = useI18n()

// Usage
t('auth.sign_in')    // → "Sign In" or "تسجيل الدخول"
t('app.name')         // → "Growa Qatar" or localized equivalent
```

Key Convention

Dot-notation keys organized by domain:

- `auth.*` — Authentication strings
- `app.*` — Application branding
- `common.*` — Shared UI labels

- `dashboard.*` — Dashboard-specific strings

Fallback: if key missing, the key itself is displayed (aids development).

RTL Support

When Arabic is selected:

- `document.documentElement.dir` set to `rtl`
 - `document.documentElement.lang` set to `ar`
 - Layout components adapt via CSS logical properties and Tailwind direction utilities
-

Bilingual Data Model

Operational entities store bilingual content:

- `name_en` / `name_ar` on farms, organizations, crops
- `description_en` / `description_ar` where applicable

UI displays content in the user's selected locale with fallback to English.

Configuration

Environment-driven defaults:

- `NEXT_PUBLIC_DEFAULT_LOCALE` — Default locale (e.g., `en`)
- `NEXT_PUBLIC_SUPPORTED_LOCALES` — Comma-separated list (e.g., `en, ar`)

Country deployment config (`lib/config/country-qatar.ts`) defines Qatar-specific locale preferences.

Coverage Scope

Area	Bilingual
Authentication pages	✔ Full
Dashboard navigation	✔ Full
Common UI components	✔ Full
Intelligence workspaces	Partial (metrics in English)
Growa AI briefings	English only (by design)
Database content	Bilingual fields where applicable

Related Notes

- 02-Platform-Architecture
- 03-Technology-Stack
- 09-Data-Model

API Design — Backend-for-Frontend Pattern

Overview

Growa uses Next.js API routes as a **Backend-for-Frontend (BFF)** layer. All external service calls and privileged operations execute server-side. The browser never receives provider API keys.

API Categories

Operations APIs

Endpoint Pattern	Methods	Purpose
/api/operations/farms	GET, POST	Farm registry management
/api/operations/custom-map-points	GET, POST	Map marker CRUD
/api/operations/custom-point-polygons	GET, POST	Polygon overlay management
/api/operations/farm-crop-insights	GET, POST	Per-point crop analytics
/api/operations/crop-types	GET, POST	Custom crop type catalog

Weather APIs (Proxied)

Endpoint Pattern	Methods	Purpose
/api/weather/by-coordinates	GET	Current weather for lat/lng
/api/weather/history/by-coordinates	GET	Historical weather data
/api/weather/grid	GET	National grid weather cells

AI APIs

Endpoint Pattern	Methods	Purpose
/api/ai/growa/analyze	POST	Server-side Growa AI briefing

Authentication Pattern

All API routes:

1. Validate session from secure cookies
 2. Reject unauthenticated requests (401)
 3. Rely on database RLS for data scoping
 4. Return only authorized data
-

Request/Response Conventions

- JSON request and response bodies
 - Standard HTTP status codes (200, 201, 400, 401, 403, 404, 500)
 - Error responses include safe error messages (no stack traces in production)
 - Input validation via Zod schemas where applicable
-

Server-Side Responsibilities

Responsibility	Layer
Session validation	API route middleware
Permission checks	API route + application hooks
Data scoping	Database RLS
External API calls	API route (keys in env only)
AI prompt assembly	Server-only AI module
Audit logging	Server-side on sensitive operations

Security Boundaries

CLIENT (Browser)

- Public env vars only
- Session cookies (HTTP-only)
- No API keys

SERVER (Next.js API Routes)

- All provider API keys
- AI prompt construction
- Privileged database operations

DATABASE (PostgreSQL + RLS)

- Final data access enforcement

Related Notes

- [12-Growa-AI-Methodology](#)
- [11-Row-Level-Security](#)
- [16-Security-Redaction-Policy](#)

IP Claims Summary

Purpose

This note summarizes the proprietary intellectual property embodied in the Growa platform, suitable for IP deposit, trade secret classification, or patent portfolio planning.

Category 1: Platform Architecture

Claim: Multi-Country Sovereign Deployment Model

A three-layer hierarchy (Platform → Country Deployment → Organization) enabling reusable codebase with sovereign data isolation per country instance.

Novel elements:

- Country-agnostic base with deployment-specific customization layer
 - Isolated database per sovereign deployment
 - Shared operational ontology across countries
-

Category 2: Module Registry System

Claim: Declarative Government Workspace Ontology

A code-first catalog defining workspace modules with role-aware labels, visibility scopes, shared data layer requirements, and navigation blueprints.

Novel elements:

- Role-specific label and submenu overrides within single module definitions
 - Six-dimensional visibility scope (org type + permissions + shared layers)
 - Progressive implementation pattern (shell UI before backend wiring)
 - Ministry role menu blueprints with filtered resolution algorithm
-

Category 3: Map-Centric UX

Claim: Persistent Map Shell with Bidirectional Data-Map Linking

An analytical dashboard where the satellite map remains mounted while workspace panels slide over it, with every data table row linked to geographic coordinates.

Novel elements:

- Map-as-operating-surface (not map-as-widget)
 - Role-aware map modes from single map instance
 - Custom polygon overlays with embedded resource metrics and scores
-

Category 4: Resource Intelligence Pipeline

Claim: Cross-Resource Agricultural Performance Analytics

A metrics pipeline from map polygons through crop insights to national KPIs, producer rankings, and at-risk identification across water, energy, and production dimensions.

Novel elements:

- Water intensity (m^3/t) and energy intensity (kWh/t) per crop and per producer
 - Irrigation pressure as water share of total resources
 - Unified efficiency scoring across resource dimensions
 - National crop matrix with share percentages
-

Category 5: Grounded Government AI

Claim: Evidence-Based Government Briefing System

A structured AI pipeline that pre-computes an operational digest from live platform data before LLM invocation, with anti-hallucination guardrails and fixed executive output schema.

Novel elements:

- Operational digest as single source of truth for LLM
- Module-specific analytical frameworks and preset prompts
- Five-section fixed briefing format (Summary → Evidence → Risks → Actions → Gaps)

- Explicit data gap transparency as first-class output
- No autonomous decision-making; human authority preserved

Category 6: Layered Permission Model

Claim: Six-Layer Permission Resolution

Effective permissions computed from country instance, organization, department, geographic scope, object scope, and action permissions — never flattened to a single role string.

Novel elements:

- Shared data layer visibility (regulatory, commercial, finance, technical)
- Four visibility levels (FULL, SUMMARY, APPROVAL, NO)
- Database RPC functions for complex authorization resolution
- Dual navigation resolution (code registry + database menus)

Category 7: Bilingual Sovereign Platform

Claim: RTL/LTR Agricultural Government Platform

Full bilingual operation with automatic layout direction switching, bilingual data model fields, and country-specific locale configuration.

Trade Secret vs. Publishable

Asset	Classification
Module registry definitions	Publishable (this vault)
Growa AI prompt templates	Trade secret
Operational digest algorithms	Publishable (conceptual)
RLS policy predicates	Confidential (excluded)
API keys and credentials	Confidential (excluded)
Role-permission exact mappings	Trade secret
Weather grid generation logic	Trade secret

Related Notes

- [04-Module-Registry](#)
- [12-Growa-AI-Methodology](#)
- [10-Auth-Access-Model](#)
- [16-Security-Redaction-Policy](#)

Security Redaction Policy

Purpose

This document records what was **deliberately excluded** from the IP Vault and why. It ensures the deposit establishes intellectual property without creating security vulnerabilities or exposing confidential business information.

Excluded: Credentials & Secrets

Item	Reason
API keys (AI, weather, database)	Direct security risk if disclosed
Service role keys	Full database bypass capability
Public/private key pairs	Authentication compromise
Environment variable values	Operational security
Hardcoded fallback credentials in source	Attack surface

Safe alternative documented: "Secrets injected at deployment time via environment configuration."

Excluded: Infrastructure Identifiers

Item	Reason
Production database hostnames	Targeted attack vector
Project IDs and instance names	Infrastructure reconnaissance
Deployment URLs (preview, staging, production)	Unauthorized access attempts
CDN and storage bucket names	Data exfiltration risk
Internal CI/CD pipeline details	Supply chain attack surface

Safe alternative documented: "Managed backend service with isolated per-deployment instances."

Excluded: Security Implementation Details

Item	Reason
Exact RLS policy SQL predicates	Policy bypass research
RPC function source code	Authorization logic reverse-engineering
Token hashing algorithms and parameters	Invitation/reset token attacks
Rate limiting thresholds	Brute force optimization
IP allowlist CIDR ranges	Network access mapping
Session token expiry exact values	Session hijacking timing

Safe alternative documented: Conceptual security strategy (default deny, membership-based, server-side enforcement).

Excluded: Internal Admin Tooling

Item	Reason
Admin email domain patterns	Privilege escalation targeting
Impersonation RPC mechanics	Unauthorized role assumption
Impersonation state persistence	Session manipulation
Debug and QA backdoors	Unauthorized access

Safe alternative documented: "Internal quality assurance tooling exists; details confidential."

Excluded: Partner & Operational Data

Item	Reason
Seeded organization names and UUIDs	Deployment-specific intelligence
Real farm and producer names	Privacy and commercial sensitivity
Live production metrics	Operational intelligence
External reference URLs in records	Third-party relationship mapping
Specific ministry personnel roles	Organizational structure intelligence

Safe alternative documented: Generic organization type examples and illustrative role templates.

Excluded: Exact Permission Matrices

Item	Reason
Role-to-permission flag exact mappings	Privilege escalation planning
Organization type visibility tables	Access boundary mapping
Shared layer assignment rules	Data layer bypass research

Safe alternative documented: Permission model architecture and flag definitions without exact mappings.

Redaction Verification Checklist

Before any IP document is published or deposited:

- ☐ No API keys, tokens, or passwords present
 - ☐ No production URLs or hostnames
 - ☐ No exact RLS or RPC implementation code
 - ☐ No internal admin domain references
 - ☐ No real organization or personnel identifiers
 - ☐ No live operational data or metrics
 - ☐ No infrastructure topology details
 - ☐ Conceptual descriptions used for all security mechanisms
-

Related Notes

- [11-Row-Level-Security](#)
- [10-Auth-Access-Model](#)
- [00-Index](#)

Glossary

Platform Terms

Term	Definition
Growa	Multi-country agricultural operations platform (base)
Growa Qatar	First sovereign country deployment for the State of Qatar
Country Instance	Top-level deployment container for a sovereign nation
Country Deployment	Customized platform instance for a specific country
Sovereign Platform	National system where data belongs to the deploying state

Organizational Terms

Term	Definition
Organization	Authorized entity operating within a country deployment
Organization Type	Classification: government_master, government, farm_company, private, public
Department	Subdivision within an organization
Membership	Link between a user and an organization with role assignment
Tenant	Organization-scoped data boundary

Access Control Terms

Term	Definition
Role Template	Predefined permission set assigned to users
Permission Flag	Application-level capability (canView, canEdit, etc.)
Scope Assignment	Geographic or object-level access boundary
Shared Data Layer	Cross-cutting visibility domain (regulatory, commercial, finance, technical)
Visibility Level	Data access granularity: FULL, SUMMARY, APPROVAL, NO
RLS	Row Level Security — database-enforced data isolation
BFF	Backend-for-Frontend — server-side API proxy pattern

Product Terms

Term	Definition
Module	A workspace capability in the platform (e.g., Live Map, Water Intelligence)
Module Registry	Declarative catalog of all platform modules
Workspace	Analytical panel that slides over the map shell
Map Shell	Persistent satellite map underlying all workspaces
Map Point	User-placed marker on the operational map
Polygon	Drawn operational area with linked metrics
Crop Insight	Per-point analytical record (production, water, energy)

Intelligence Terms

Term	Definition
Water Intensity	Cubic metres of water per ton of crop production (m³/t)
Energy Intensity	Kilowatt-hours per ton of production (kWh/t)
Irrigation Pressure	Water's share of total resource consumption
Efficiency Score	Composite performance metric for producers/areas
At-Risk Producer	Farm flagged for low efficiency or high resource pressure
Operational Digest	Pre-computed data summary fed to Growa AI
Growa AI	Grounded government briefing system

Technical Terms

Term	Definition
i18n	Internationalization — multi-language support
RTL	Right-to-left text direction (Arabic)
LTR	Left-to-right text direction (English)
SSR	Server-Side Rendering
App Router	Next.js file-based routing system
Migration	Versioned database schema change file
RPC	Remote Procedure Call — database function invoked from application

Compliance Terms

Term	Definition
Non-Conformity	Documented violation of agricultural standards
Corrective Action	Required remediation following a non-conformity
Compliance Case	End-to-end regulatory case with evidence trail
Inspection	Field verification of agricultural compliance
Evidence	Documentary proof attached to inspections or cases

[← Back to 00-Index](#)